

(TR-104) PYTHON WITH AI/ML

Training Day 15 Report

Date: 30 January 2026

Topic Covered: Online Payment Fraud Detection System – Python Project using Pandas & Matplotlib

The fifteenth day of training focused on developing an Online Payment Fraud Detection System using Python. The project involved analyzing transaction data and identifying suspicious activities using statistical techniques and data-driven logic.

- Imported the dataset and converted it into a structured Pandas DataFrame for analysis.
- Performed data cleaning and exploratory data analysis to understand transaction patterns.
- Calculated statistical values such as mean and standard deviation for numerical insights.
- Developed a rule-based risk classification system to categorize transactions into Low Risk, Medium Risk, and High Risk.
- Created additional derived features to improve fraud detection accuracy.
- Structured the project into modular components including data loading, analysis, risk modeling, and visualization.
- Generated visual insights using Matplotlib, including histograms and bar charts.
- Analyzed transaction distributions and identified potential fraud patterns through visualization.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/Fraud_Detection
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/online-payment-fraud-detection-using-python-62106c4722a1>